

THE UNIVERSITY of EDINBURGH

Edinburgh Research Explorer

Trait-dependent associations between early- and late-life reproduction in a wild mammal

Citation for published version:
McKenna-Ell, C, Ravindran, S, Pilkington, JG, Pemberton, JM, Nussey, DH & Froy, H 2023, 'Trait-dependent associations between early- and late-life reproduction in a wild mammal', *Biology letters*, vol. 19, no. 7, 20230050. https://doi.org/10.1098/rsbl.2023.0050

Digital Object Identifier (DOI):
10.1098/rsbl.2023.0050

Link:
Link to publication record in Edinburgh Research Explorer

Document Version:
Publisher's PDF, also known as Version of record

Published In:
Biology letters

General rights
Copyright for the publications made accessible via the Edinburgh Research Explorer is retained by the author(s) and / or other copyright owners and it is a condition of accessing these publications that users recognise and abide by the legal requirements associated with these rights.

Take down policy
The University of Edinburgh has made every reasonable effort to ensure that Edinburgh Research Explorer content complies with UK legislation. If you believe that the public display of this file breaches copyright please contact openaccess@ed.ac.uk providing details, and we will remove access to the work immediately and investigate your claim.



Download date: 27. Oct. 2026

BIOLOGY LETTERS

royalsocietypublishing.org/journal/rsbl

Research

Cite this article: McKenna-Ell C, Ravindran S, Pilkington JG, Pemberton JM, Nussey DH, Froy H. 2023 Trait-dependent associations between early- and late-life reproduction in a wild mammal. *Biol. Lett.* **19**: 20230050. https://doi.org/10.1098/rsbl.2023.0050

Received: 2 February 2023
Accepted: 23 June 2023

Subject Category:
Evolutionary biology

Subject Areas:
ecology, evolution

Keywords:
senescence, ageing, life-history trade-off, Soay sheep *Ovis aries*

Author for correspondence:
Hannah Froy
e-mail: hannah.froy@ed.ac.uk

Electronic supplementary material is available online https://doi.org/10.6084/m9.figshare.c.6727604.

THE ROYAL SOCIETY PUBLISHING

Trait-dependent associations between early- and late-life reproduction in a wild mammal

Chris McKenna-Ell, Sanjana Ravindran, Jill G. Pilkington, Josephine M. Pemberton, Daniel H. Nussey and Hannah Froy

Institute of Ecology and Evolution, School of Biological Sciences, University of Edinburgh, Edinburgh EH9 3FL, UK

SR, 0000-0003-0996-0262; JMP, 0000-0002-0075-1504; DHN, 0000-0002-9985-0317; HF, 0000-0003-2965-3526

Early- versus late-life trade-offs are a central prediction of life-history theory that are expected to shape the evolution of ageing. While ageing is widely observed in wild vertebrates, evidence that early–late trade-offs influence ageing rates remains limited. Vertebrate reproduction is a complex, multi-stage process, yet few studies have examined how different aspects of early-life reproductive allocation shape late-life performance and ageing. Here, we use longitudinal data from a 36-year study of wild Soay sheep to show that early-life reproduction predicts late-life reproductive performance in a trait-dependent manner. Females that started breeding earlier showed more rapid declines in annual breeding probability with age, consistent with a trade-off. However, age-related declines in offspring first-year survival and birth weight were not associated with early-life reproduction. Selective disappearance was evident in all three late-life reproductive measures, with longer-lived females having higher average performance. Our results provide mixed support for early–late reproductive trade-offs and show that the way early-life reproduction shapes late-life performance and ageing can differ among reproductive traits.

1. Introduction

Senescence, the age-related decline in physiological function and reproductive value, is prevalent across the tree of life [1]. Despite its ubiquity, senescence is a hugely diverse process that varies across many biological levels: among cells, tissues and organs, through different phenotypic traits, to variation between individuals, sexes, populations and species [2–5]. Trade-offs between early-life reproduction and the maintenance of physiological and reproductive function in later adulthood form a cornerstone of the antagonistic pleiotropy and disposable soma theories on the evolution of ageing [6–8]. This theory predicts that investment of limited resources in energetically demanding activities such as growth and reproduction may come at a cost in terms of somatic maintenance [9,10]. Variation in allocation to reproduction in early life is therefore expected to be a key driver of among-individual variation in the onset and rate of senescence [6,11].

In vertebrates, reproduction is a composite trait which can be conceptualized sequentially, from the probability of breeding, successful mating, and fertilization to offspring production, provisioning and successful recruitment to the breeding population [12]. While there is widespread evidence that reproductive traits show senescent declines in wild vertebrates [13,14], there is also increasing evidence of asynchrony of senescence among reproductive traits within populations [3,15,16]. While early–late trade-offs are well documented in laboratory model organisms [17], evidence from natural systems remains mixed [18–21] as heterogeneity among individuals in resource acquisition and selective disappearance can mask both trade-offs and ageing patterns [22–24]. Recent studies confirm that

© 2023 The Authors. Published by the Royal Society under the terms of the Creative Commons Attribution License http://creativecommons.org/licenses/by/4.0/, which permits unrestricted use, provided the original author and source are credited.

Downloaded from http://royalsocietypublishing.org/rsbl/article-pdf/doi/10.1098/rsbl.2023.0050/442314/rsbl.2023.0050.pdf
by University of Edinburgh user
on 25 February 2026

where longitudinal lifelong data are available and appropriate statistical controls are used to account for heterogeneity, early–late trade-offs can be detected in wild populations [11,25,26]. Yet, few studies have tested whether and how trade-offs between early- and late-life reproduction might differ across reproductive traits.

In this study, we used 36-years of longitudinal data from an unmanaged population of Soay sheep on St Kilda, Scotland, to test whether reproduction in early life was associated with increased rates of senescence in three components of female reproduction: breeding probability, offspring birth weight and offspring first-winter survival, thereby testing a core prediction of some of the key evolutionary theories of ageing in the wild.

2. Methods

(a) Study system

Longitudinal monitoring of Soay sheep (*Ovis aries*) on the island of Hirta in the St Kilda archipelago, Scotland, has been ongoing since 1985. Ewes conceive during the annual autumn rut and gestate over the winter months, giving birth to singleton or twin lambs in late March–April. Ninety-five per cent of lambs born in the study are caught, sampled and uniquely tagged shortly after birth [27]. Parentage is determined through a combination of field observation and genetic methods [27]. Females sexually mature during their first year, and 44% of those that survive to 1 March in the year following birth give birth to a live lamb around their first birthday. The vast majority of mortality occurs in late winter and early spring (89% deaths occur January–April). Mortality searches during this period in combination with 30 censuses conducted throughout the year mean that the fate of individuals is known with a high degree of accuracy. Senescent declines have been documented in morphological, physiological and demographic traits including components of female reproductive performance [3,28,29]. In this study, we used data on known aged females collected between 1985 and 2020.

(b) Reproduction in early and later life

We considered three key components of annual reproductive performance in later life:

— *Breeding probability (0/1)*: probability of giving birth to a live lamb conditional on a female being alive on 1 March year t ($n = 3173$ observations of 762 females)
— *Offspring first-winter survival (0/1)*: probability of a female having at least one surviving yearling on 1 May year $t + 1$ conditional on giving birth to a live lamb in year t ($n = 2573$ of 714)
— *Offspring birth weight*: weight of lamb on first capture to the nearest 0.1 kg, corrected for age at capture (in days) ($n = 2317$ of 649).

We did not include twinning probability, the other major component of female reproductive performance, because the rate of twinning is low (approx. 11%), and it does not show signs of senescence (electronic supplementary material, figure S1). Based on the population-level ageing trajectories of our chosen traits (electronic supplementary material, figure S1), we designated ages 5 years and older as 'later life'. This captured the peak performance for all three traits while minimizing nonlinearity in the subsequent decline. We re-ran our analyses using cut-offs at age 4 and 6 to check that our results were not dependent on the precise threshold used.

We used two traits to capture variation in early-life reproduction:

— *Bred as a yearling (0/1)*: whether a female born in year t gave birth to a live lamb in year $t + 1$. Note that females in this age class never have twins.
— *Early-life recruitment*: the number of offspring a female successfully raised through their first winter (to 1 May the year following birth) in the first 4 years of her life (i.e. lambs born when she was aged 1–3 years).

The distribution of early-life traits for individuals included in the analyses of later life reproduction are shown in the electronic supplementary material, figure S2. There was little overlap between the traits, with females that bred as yearlings having a similar distribution of early-life recruitment compared to those that did not (electronic supplementary material, figure S2).

(c) Statistical analysis

We modelled reproductive performance in later life as a function of early-life reproduction using (generalized) linear mixed-effects models. Breeding probability and offspring first-winter survival were modelled as binary traits with a binomial error distribution and logit link function. Offspring birth weight was modelled as a Gaussian trait with identity link function. We included a linear effect of age in all models to capture senescence in reproductive performance beyond the age of 5. Age at last observation was included as a fixed covariate to account for the selective disappearance of shorter-lived individuals that may differ in their performance throughout later life [23]. We used age at last observation rather than longevity so we could include individuals that were still alive in our analyses (less than 10% of observations). The inclusion of this term meant that we could interpret the effect of age as a within-individual effect in our models. To test for associations between early-life reproduction and average performance in later life, we included whether a female bred as a yearling as a two-level fixed factor and early-life recruitment as a fixed covariate. We also included each of these terms in a two-way interaction with age to test whether the rate of ageing was dependent on reproduction in early life. In our models of offspring birth weight, we included offspring age at time of capture in days as a fixed covariate to account for the rapid growth that occurs during this period, offspring sex and whether they were a twin as two-level fixed factors as these factors are known to influence birth weight. All models included female identity and observation year as random intercept terms to account for repeated measures of individuals and years. We also included female cohort as a random intercept term to account for the fact that females born in the same year experienced similar conditions during early life. All terms were retained in the model regardless of statistical significance. Models were run using *lme4* [30] in the R programming environment (v. 4.2.2; [31]).

3. Results

Breeding probability, offspring first-winter survival and offspring birth weight all declined with increasing age in females aged 5 and older (table 1; electronic supplementary material, figure S3). Selective disappearance was evident in all three reproductive traits with females that were older at last observation having higher average trait values (table 1; electronic supplementary material, figure S3).

Reproduction in early life was positively associated with later life breeding probability: females who bred as yearlings and those who successfully recruited more offspring in their

Downloaded from http://royalsocietypublishing.org/rsbl/article-pdf/doi/10.1098/rsbl.2023.0050/442314/rsbl.2023.0050.pdf
by University of Edinburgh user
on 25 February 2026

Table 1. Fixed and random effect estimates from (generalized) linear mixed-effects models of (A) breeding probability; (B) offspring first-winter survival and (C) offspring birth weight in female Soay sheep aged 5 years and older. Breeding probability and offspring survival were modelled using a binomial error distribution and logit link function; offspring birth weight was Gaussian with an identity link function. Estimates are given on the latent scale. Fixed effects where $p < 0.05$ are highlighted in italics.

random effects	variance	fixed effects	estimate	s.e.	*p*-value
(A) breeding probability					
female identity	3.061	intercept	3.287	0.549	<0.001***
year	0.514	*age*	*−0.487*	*0.060*	*<0.001****
female cohort	0.147	*age at last observation*	*0.188*	*0.054*	*<0.001****
		bred as yearling	*2.077*	*0.587*	*<0.001****
		early-life recruitment	*0.948*	*0.464*	*0.041**
		age : bred as yearling	*−0.210*	*0.075*	*0.005***
n = 3173 observations of 762 females		age : early-life recruitment	0.002	0.058	0.972
(B) offspring first-winter survival					
female identity	0.207	intercept	0.077	0.442	0.862
offspring birth year	1.491	*age*	*−0.232*	*0.054*	*<0.001****
female cohort	0.004	*age at last observation*	*0.089*	*0.032*	*0.006***
		bred as yearling	−0.468	0.415	0.260
		early-life recruitment	*0.869*	*0.309*	*0.005***
		age : bred as yearling	0.038	0.060	0.527
n = 2573 observations of 714 females		age : early-life recruitment	−0.063	0.043	0.149
(C) offspring birth weight					
female identity	0.066	intercept	3.013	0.097	<0.001***
offspring birth year	0.057	*offspring capture age*	*0.112*	*0.003*	*<0.001****
female cohort	0.004	*offspring sex*	*0.106*	*0.016*	*<0.001****
residual	0.115	*offspring twin status*	*−0.826*	*0.020*	*<0.001****
		age	*−0.061*	*0.009*	*<0.001****
		age at last observation	*0.016*	*0.007*	*0.031**
		bred as yearling	0.053	0.070	0.443
		early-life recruitment	0.019	0.050	0.705
		age : bred as yearling	0.005	0.010	0.572
n = 2317 observations of 649 females		age : early-life recruitment	0.005	0.007	0.432

first 4 years of life were more likely to breed annually from 5 years of age (table 1A; electronic supplementary material, figure S3A). The rate of senescence in breeding probability was associated with early-life reproduction: females who bred as yearlings suffered more rapid declines in breeding probability than those that did not (interaction between female age and bred as yearling = -0.210 ± 0.075, $p = 0.005$; table 1A; figure 1*a*). By contrast, successfully raising more offspring to recruitment was not associated with more rapid ageing in breeding probability (table 1A; figure 1*d*).

Females with higher early-life recruitment success also had higher offspring survival in later life, but there was no association between breeding as a yearling and late-life offspring survival (table 1B; electronic supplementary material, figure S3B). There was no association between reproduction in early life and offspring birth weight in later life (table 1C; electronic supplementary material, figure S3C). The rate of ageing in offspring first-winter survival and offspring birth weight was not dependent on whether a female bred as a yearling or their recruitment success during early life (table 1B,1C: figure 1*b*,*c*,*ef*). Our results were not dependent on the age threshold used to define later life: excluding observations from females below 4 or 6 years of age did not qualitatively change our results (electronic supplementary material, table S1).

4. Discussion

We found mixed evidence that early–late trade-offs drive variation in the rate of reproductive senescence in female Soay sheep. Females who bred as yearlings had a higher probability of breeding at the age of 5 years, but this was associated with a faster rate of age-related decline in breeding probability (as indicated by the crossed lines in figure 1*a*). This suggests a trade-off in performance between early and later life and supports key theoretical predictions that among-individual variation in the rate of ageing may be driven by differences in early-life reproductive allocation. However, breeding as a yearling was not associated with

Downloaded from http://royalsocietypublishing.org/rsbl/article-pdf/doi/10.1098/rsbl.2023.0050/442314/rsbl.2023.0050.pdf
by University of Edinburgh user
on 25 February 2026

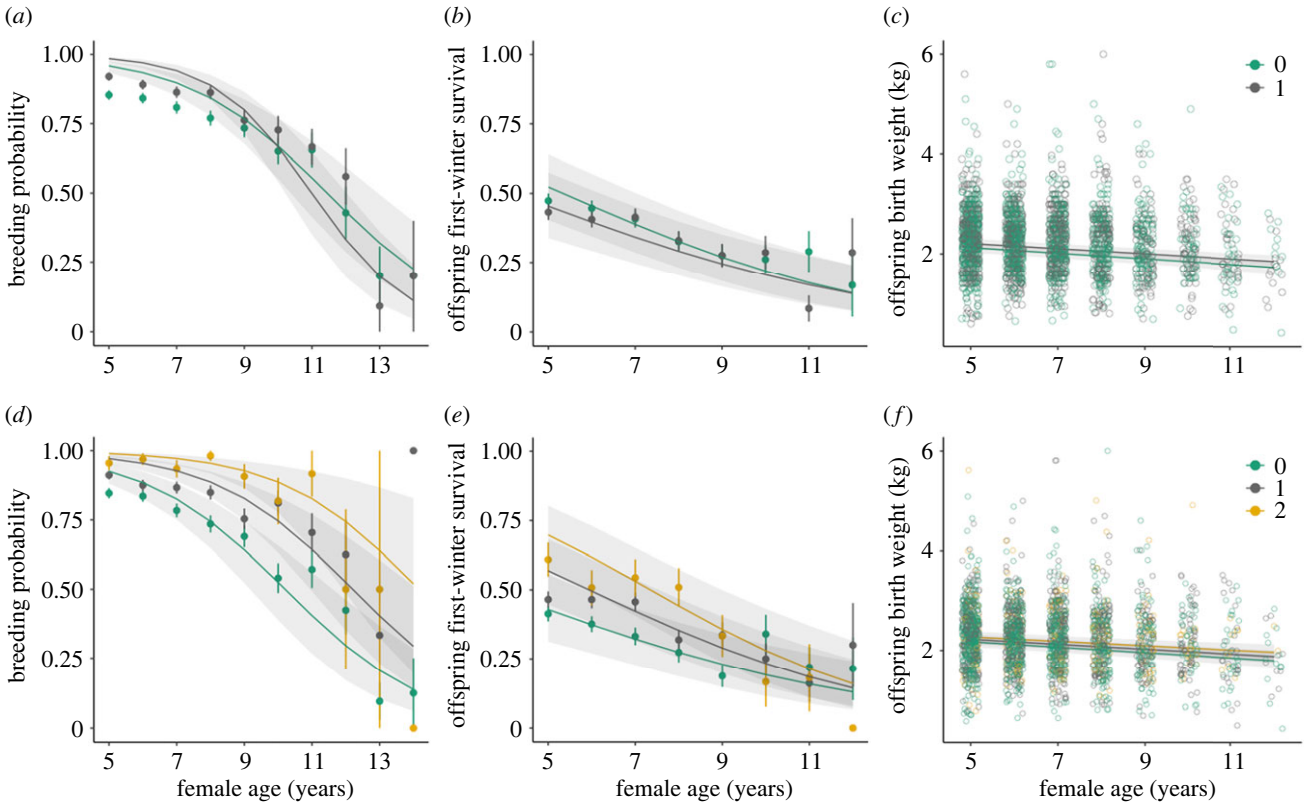


Figure 1. Associations between early-life reproduction (breeding as a yearling and early-life recruitment) and age-specific reproduction in later life in Soay sheep. Predictions and 95% confidence intervals from (generalized) linear mixed-effects models of (*a*,*d*) female breeding probability; (*b*,*e*) offspring first-winter survival and (*c*,*f*) offspring birth weight in female sheep aged 5 years and older. Plots show the effect of female or maternal age along with the additive or interactive effects of early-life reproduction. Plots (*a*–*c*) show the effect of breeding as a yearling (grey) versus not (green); plots (*d*–*f*) show the effect of successfully recruiting 0, 1 or 2 offspring in the first 4 years of life (0 = green; 1 = grey; 2 = yellow). Lines show the model predictions with 95% confidence intervals in shaded grey; points show the raw data means and standard errors (plots (*a*), (*b*), (*d*) and (*e*)) or raw data (plots (*c*) and (*f*)). Full models are shown in table 1.

the rate of senescence in offspring first-winter survival or offspring birth weight (figure 1*b*,*c*). Similarly, the number of offspring that a female managed to successfully recruit during early life was not associated with the rate of senescence in any of our reproductive traits (figure 1*d*,*e*,*f*). Overall, our results therefore suggest that reproduction in early life is not a key driver of among-individual variation in reproductive senescence in this system.

In line with previous findings, there was evidence for senescence in multiple components of female reproduction (table 1) [3,28]. These declines were substantial: the probability of giving birth to a live lamb declined from a mean of 0.96 (95%CI 0.93–0.97) aged 5 to 0.32 (0.18–0.50) aged 13 years; the probability of such a lamb surviving its first winter declined from 0.52 (0.40–0.64) to 0.11 (0.06–0.20), and mean singleton lamb weight declined from 2.21 kg (2.11–2.31) to 1.80 kg (1.66–1.93). Senescent declines were therefore apparent at multiple stages of reproduction, from successful oocyte production and gestation through to lactation and offspring provisioning. Life-history theory of ageing predicts that these declines may reflect a delayed cost of investment in reproduction, with females that breed successfully in the first few years of life allocating relatively less to somatic maintenance and consequently suffering more rapid senescence [9]. This pattern has been observed in some wild vertebrate study systems [20,25,26], but not in others [18,19,21]. Our results suggest that the detectability of long-term costs of early-life reproductive allocation may depend on the reproductive trait being investigated, which may help explain the mixed evidence from field studies to date. Furthermore, there is mounting evidence for asynchrony in the patterns of senescence across reproductive traits in wild vertebrates [3,32,33,34]. Variation in the manifestation of early versus late trade-offs among reproductive traits may help explain some of this asynchrony.

Overall, our metrics of early-life reproduction tended to be positively associated with average reproductive performance in later life (table 1). This suggests heterogeneity among females that persists over the lifetime, potentially masking trade-offs between early and later life [22,35]. Such 'carry over' or 'quality' effects are widely observed in wild systems and may be driven by genetics, persistent early-life effects, or aspects of an individual's environment that remain constant throughout life such as home range quality [17,36]. Such heterogeneity was also evident as selective disappearance in all three traits in later life (table 1), further highlighting the importance of longitudinal data and statistically controlling for such effects in the study of reproductive ageing [23,24]. Short-term reproduction–survival trade-offs may also play an important role in our failure to observe predicted early versus late reproductive trade-offs, if individuals that invest heavily in early-life reproduction are less likely to survive to senescent ages. Indeed, reproduction in yearlings is strongly environment-dependent and associated with reduced yearling survival in this population [37]. Our analyses included only females that survived to age 5 and so would not capture short-term survival costs of early reproduction. Overall, our results do not support an important role

Downloaded from http://royalsocietypublishing.org/rsbl/article-pdf/doi/10.1098/rsbl.2023.0050/442314/rsbl.2023.0050.pdf
by University of Edinburgh user
on 25 February 2026

for early–late reproductive trade-offs in driving variation in reproductive ageing in wild Soay sheep. However, they do highlight the need for increased consideration of different reproductive traits in studies of reproductive ageing and early–late trade-offs, and the importance of understanding the drivers of among-individual heterogeneity in adult reproductive performance in wild populations.

Ethics. This research was conducted under UK Home Office license, most recently PP4825594.

Data accessibility. Data and accompanying description are available from the Dryad Digital Repository: https://doi.org/10.5061/dryad.stqjq2c7s [38].

The data are provided in the electronic supplementary material [39].

Authors' contributions. C.M.-E.: conceptualization, formal analysis, methodology and writing—review and editing; S.R.: conceptualization, data curation, methodology, supervision and writing—review and editing; J.G.P.: data curation, methodology and writing—review and editing; J.M.P.: funding acquisition, methodology, project administration and writing—review and editing; D.H.N.: conceptualization, funding acquisition, project administration, supervision, writing—original draft and writing—review and editing; H.F.: conceptualization, data curation, formal analysis, methodology, project administration, supervision, visualization, writing—original draft and writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. We declare we have no competing interests.

Funding. This work was supported by funding from the Natural Environment Research Council, Biotechnology and Biological Sciences Research Council, European Research Council and the Royal Society.

Acknowledgements. We are indebted to all those involved in the study of Soay sheep on St Kilda over many years, and thank the National Trust for Scotland for permission to work on St Kilda.

References

1. Shefferson RP, Jones OR, Salguero-Gómez R. 2017 *The evolution of senescence in the tree of life*. Cambridge, UK: Cambridge University Press.
2. Rando TA, Wyss-Coray T. 2021 Asynchronous, contagious and digital aging. *Nat. Aging* **1**, 29–35. (doi:10.1038/s43587-020-00015-1)
3. Hayward AD, Moorad J, Regan CE, Berenos C, Pilkington JG, Pemberton JM, Nussey DH. 2015 Asynchrony of senescence among phenotypic traits in a wild mammal population. *Exp. Gerontol.* **71**, 56–68. (doi:10.1016/j.exger.2015.08.003)
4. Cayuela H *et al.* 2021 Thermal conditions predict intraspecific variation in senescence rate in frogs and toads. *Proc. Natl Acad. Sci. USA* **118**, e2112235118. (doi:10.1073/pnas.2112235118)
5. Jones OR *et al.* 2014 Diversity of ageing across the tree of life. *Nature* **505**, 169–173. (doi:10.1038/nature12789)
6. Williams GC. 1957 Pleiotropy, natural selection, and the evolution of senescence. *Evolution* **11**, 398–411. (doi:10.2307/2406060)
7. Kirkwood TBL. 1977 Evolution of ageing. *Nature* **270**, 301–304. (doi:10.1038/270301a0)
8. Maklakov AA, Chapman T. 2019 Evolution of ageing as a tangle of trade-offs: energy versus function. *Proc. R. Soc. B* **286**, 20191604. (doi:10.1098/rspb.2019.1604)
9. Kirkwood TB, Rose MR. 1991 Evolution of senescence: late survival sacrificed for reproduction. *Phil. Trans. R. Soc. Lond. Ser. B* **332**, 15–24. (doi:10.1098/rstb.1991.0028)
10. Cody ML. 1966 A general theory of clutch size. *Evolution* **20**, 174–184. (doi:10.2307/2406571)
11. Lemaître JF, Berger V, Bonenfant C, Douhard M, Gamelon M, Plard F, Gaillard JM. 2015 Early-late life trade-offs and the evolution of ageing in the wild. *Proc. R. Soc. B* **282**, 20150209. (doi:10.1098/rspb.2015.0209)
12. Clutton-Brock TH. 1988 *Reproductive success: studies of individual variation in contrasting breeding systems*. Chicago, IL: University of Chicago Press.
13. Nussey DH, Froy H, Lemaitre JF, Gaillard JM, Austad SN. 2013 Senescence in natural populations of animals: widespread evidence and its implications for bio-gerontology. *Ageing Res. Rev.* **12**, 214–225. (doi:10.1016/j.arr.2012.07.004)
14. Lemaître JF, Gaillard JM. 2017 Reproductive senescence: new perspectives in the wild. *Biol. Rev.* **92**, 2182–2199. (doi:10.1111/brv.12328)
15. Massot M, Clobert J, Montes-Poloni L, Haussy C, Cubo J, Meylan S. 2011 An integrative study of ageing in a wild population of common lizards. *Funct. Ecol.* **25**, 848–858. (doi:10.1111/j.1365-2435.2011.01837.x)
16. Lecomte VJ *et al.* 2010 Patterns of aging in the long-lived wandering albatross. *Proc. Natl Acad. Sci. USA* **107**, 6370–6375. (doi:10.1073/pnas.0911181107)
17. Austad SN, Hoffman JM. 2018 Is antagonistic pleiotropy ubiquitous in aging biology? *Evol. Med. Public Health* **2018**, 287–294. (doi:10.1093/emph/eoy033)
18. Panagakis A, Hamel S, Côté SD. 2017 Influence of early reproductive success on longevity and late reproductive success in an alpine ungulate. *Am. Nat.* **189**, 667–683. (doi:10.1086/691388)
19. Rodríguez-Muñoz R, Boonekamp JJ, Liu XP, Skicko I, Fisher DN, Hopwood P, Tregenza T. 2019 Testing the effect of early-life reproductive effort on age-related decline in a wild insect. *Evolution* **73**, 317–328. (doi:10.1111/evo.13679)
20. Nussey DH, Kruuk LEB, Morris A, Clutton-Brock TH. 2007 Environmental conditions in early life influence ageing rates in a wild population of red deer. *Curr. Biol.* **17**, R1000–R1001. (doi:10.1016/j.cub.2007.10.005)
21. Pigeon G, Landes J, Festa-Bianchet M, Pelletier F. 2021 Do early-life conditions drive variation in senescence of female bighorn sheep? *Front. Cell Dev. Biol.* **9**, 637692. (doi:10.3389/fcell.2021.637692)
22. Van Noordwijk AJ, de Jong G. 1986 Acquisition and allocation of resources: their influence on variation in life history tactics. *Am. Nat.* **128**, 137–142. (doi:10.1086/284547)
23. van de Pol M, Verhulst S. 2006 Age-dependent traits: a new statistical model to separate within- and between-individual effects. *Am. Nat.* **167**, 766–773. (doi:10.1086/503331)
24. Nussey DH, Coulson T, Festa-Bianchet M, Gaillard JM. 2008 Measuring senescence in wild animal populations: towards a longitudinal approach. *Funct. Ecol.* **22**, 393–406. (doi:10.1111/j.1365-2435.2008.01408.x)
25. Reed TE, Kruuk LEB, Wanless S, Frederiksen M, Cunningham EJA, Harris MP. 2008 Reproductive senescence in a long-lived seabird: rates of decline in late-life performance are associated with varying costs of early reproduction. *Am. Nat.* **171**, E89–E101. (doi:10.1086/524957)
26. Bouwhuis S, Charmantier A, Verhulst S, Sheldon BC. 2010 Individual variation in rates of senescence: natal origin effects and disposable soma in a wild bird population. *J. Anim. Ecol.* **79**, 1251–1261. (doi:10.1111/j.1365-2656.2010.01730.x)
27. Clutton-Brock TH, Pemberton JM. 2004 *Soay sheep: dynamics and selection in an island population*. Cambridge, UK: Cambridge University Press.
28. Hayward AD, Wilson AJ, Pilkington JG, Clutton-Brock TH, Pemberton JM, Kruuk LEB. 2013 Reproductive senescence in female Soay sheep: variation across traits and contributions of individual ageing and selective disappearance. *Funct. Ecol.* **27**, 184–195. (doi:10.1111/1365-2435.12029)
29. Froy H, Sparks A, Watt K, Sinclair R, Bach F, Pilkington J, Pemberton J, McNeilly T, Nussey D. 2019 Senescence in immunity against helminth parasites predicts adult mortality in a wild mammal. *Science* **365**, 1296–1298. (doi:10.1126/science.aaw5822)
30. Bates D, Mächler M, Bolker B, Walker S. 2015 Fitting linear mixed-effects models using lme4. *J. Stat. Softw.* **67**, 48. (doi:10.18637/jss.v067.i01)

Downloaded from http://royalsocietypublishing.org/rsbl/article-pdf/doi/10.1098/rsbl.2023.0050/442314/rsbl.2023.0050.pdf
by University of Edinburgh user
on 25 February 2026

31. R Core Team. 2022 *R: a language and environment for statistical computing*. Vienna, Austria: R Foundation for Statistical Computing.
32. Evans SR, Gustafsson L, Sheldon BC. 2011 Divergent patterns of age-dependence in ornamental and reproductive traits in the collared flycatcher. *Evolution* **65**, 1623–1636. (doi:10.1111/j.1558-5646.2011.01253.x)
33. Cooper EB, Bonnet T, Osmond HL, Cockburn A, Kruuk LE. 2021 Aging and senescence across reproductive traits and survival in superb fairy-wrens (*Malurus cyaneus*). *Am. Nat.* **197**, 111–127. (doi:10.1086/711755)
34. Naciri M, Aars J, Blanchet MA, Gimenez O, Cubaynes S. 2022 Reproductive senescence in polar bears in a variable environment. *Front. Ecol. Evol.* **10**, 920481. (doi:10.3389/fevo.2022.920481)
35. Vedder O, Bouwhuis S. 2018 Heterogeneity in individual quality in birds: overall patterns and insights from a study on common terns. *Oikos* **127**, 719–727. (doi:10.1111/oik.04273)
36. Wilson AJ, Nussey DH. 2010 What is individual quality? An evolutionary perspective. *Trends Ecol. Evol.* **25**, 207–214. (doi:10.1016/j.tree.2009.10.002)
37. Tavecchia G, Coulson T, Morgan BJ, Pemberton JM, Pilkington J, Gulland F, Clutton-Brock TH. 2005 Predictors of reproductive cost in female Soay sheep. *J. Anim. Ecol.* **74**, 201–213. (doi:10.1111/j.1365-2656.2005.00916.x)
38. Froy H, McKenna-Ell C, Ravindran S, Pilkington JG, Pemberton JM, Nussey DH. 2023 Data from: Trait-dependent associations between early- and late-life reproduction in a wild mammal. Dryad Digital Repository. (doi:10.5061/dryad.stqjq2c7s)
39. McKenna-Ell C, Ravindran S, Pilkington JG, Pemberton JM, Nussey DH, Froy H. 2023 Trait-dependent associations between early- and late-life reproduction in a wild mammal. Figshare. (doi:10.6084/m9.figshare.c.6727604)

Downloaded from http://royalsocietypublishing.org/rsbl/article-pdf/doi/10.1098/rsbl.2023.0050/442314/rsbl.2023.0050.pdf
by University of Edinburgh user
on 25 February 2026